



JJ COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING

METALLURGY AND MATERIALS SCIENCE

ME25CO8 - Semester III

UNITS III & IV

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How to Use This Document in the Examination

Recommended answer presentation
1-mark MCQ: Read the complete situation, identify the property being asked, and select the most appropriate option.
3-mark answer: Write 3 clear technical points. Start with the selection/definition, then give the reason and one property or application.
12-mark answer: Use a short introduction, clear subheadings, a comparison table/flowchart where useful, application-based justification, and a brief conclusion.
Underline important keywords such as corrosion resistance, hardenability, toughness, matrix, reinforcement, aging and transformation toughening.
For material-selection questions, a table with Component - Selected material - Required property - Reason is usually the clearest format.
For treatment/process questions, write the sequence first as a flowchart, then explain what happens at each step.

Simple 12-mark writing pattern

1. Introduction/definition: 2-3 lines to show the basic concept.
2. Main answer: 5-8 technical points or a well-organized comparison table.
3. Diagram/flowchart: include whenever the question contains a process, mechanism or microstructural change.
4. Application-based justification: link the property of the material to the requirement of the component.
5. Conclusion: one or two lines stating the final selection or engineering significance.

UNIT III - FERROUS AND NON-FERROUS METALS: QUESTION BANK

Coverage: alloying additions; stainless, tool, HSLA and maraging steels; cast irons; copper alloys; aluminium and Al-Cu precipitation hardening; titanium, magnesium and Ni-based alloys; shape memory alloys; materials standards.

A. One-Mark / MCQ Practice

No.	Question	Ans.
1	A stainless-steel food-processing vessel needs corrosion resistance. Which alloying element is essential for forming a protective oxide film? (a) Mn (b) Cr (c) V (d) W	b
2	A tool must retain hardness at elevated cutting temperature. Which alloying element is especially associated with hot hardness in high-speed tool steels? (a) W (b) Cu (c) Al (d) Mg	a
3	A steel component needs improved toughness, especially at low temperature. Which alloying element is commonly useful? (a) Ni (b) S (c) P (d) Pb	a
4	A structural steel must give high strength with good weldability and reduced section thickness. Which steel class is appropriate? (a) White cast iron (b) HSLA steel (c) High-carbon tool steel (d) Grey cast iron	b
5	A very-high-strength steel is strengthened mainly by aging of a low-carbon, high-nickel martensitic matrix. Identify it. (a) Maraging steel (b) Mild steel (c) Grey cast iron (d) Brass	a
6	Which cast iron is preferred for a machine bed because graphite flakes help damping and machinability? (a) White cast iron (b) Grey cast iron (c) Malleable cast iron (d) Chilled iron	b
7	A crusher roll requires very high abrasion resistance and can tolerate brittleness. Which cast iron is suitable? (a) Grey (b) White (c) Malleable (d) Spheroidal graphite	b

No.	Question	Ans.
8	A pipe fitting requires better ductility than white cast iron. Which type is produced by heat treatment of white iron? (a) Grey (b) Malleable (c) Alloy grey (d) Compacted graphite	b
9	An automobile crankshaft needs castability with high strength and toughness. Which cast iron is most suitable? (a) White (b) Grey (c) Spheroidal graphite (d) Mottled	c
10	Which copper alloy is most suitable for seawater condenser tubes? (a) Brass (b) Cupronickel (c) Pure zinc (d) Al-Si alloy	b
11	A common bearing bush needs good wear resistance and load carrying ability. Which copper alloy is a typical choice? (a) Bronze (b) Cupronickel (c) Pure aluminium (d) Magnesium	a
12	In precipitation hardening of Al-Cu, what follows solution treatment and quenching? (a) Aging (b) Carburizing (c) Normalising (d) Spheroidising	a
13	Which non-ferrous alloy family is attractive for aircraft structures because of low density and useful strength-to-weight ratio? (a) Aluminium alloys (b) White cast iron (c) Lead alloys (d) High-carbon steel	a
14	A turbine blade operates at very high temperature under stress. Which material family is most suitable? (a) Ni-based superalloy (b) PE (c) Grey cast iron (d) Brass	a
15	A thermal actuator is required to recover a preset shape on heating. Which material is suitable? (a) Shape memory alloy (b) Tool steel (c) Bronze (d) HSLA steel	a

Quick reason key:

1. b - Chromium forms a stable passive Cr-rich oxide film.

2. a - Tungsten forms stable hard carbides and improves red/hot hardness.
3. a - Nickel improves toughness and ductility without forming brittle carbides.
4. b - HSLA steels use small alloy additions to raise strength while retaining weldability.
5. a - Maraging steel is age hardened by fine intermetallic precipitates.
6. b - Grey iron has flake graphite, good damping and machinability.
7. b - White cast iron contains hard cementite and offers excellent wear resistance.
8. b - Malleable iron is obtained by annealing white iron to form temper carbon.
9. c - Nodular graphite reduces stress concentration and improves strength and ductility.
10. b - Cu-Ni alloys have strong resistance to seawater corrosion and erosion.
11. a - Bronze is widely used for bearings due to wear resistance and anti-friction behavior.
12. a - Aging allows fine strengthening precipitates to form from the supersaturated solid solution.
13. a - Al alloys combine low density with useful structural strength.
14. a - Ni superalloys retain strength and creep resistance at high temperature.
15. a - Shape memory alloys recover shape through a reversible martensitic transformation.

B. Three-Mark Questions with Answer Guides

1. Explain why chromium and nickel are often used together in stainless steels.

Answer guide: Cr provides passivation/corrosion resistance; Ni improves toughness and stabilizes austenite; together they give corrosion resistance with useful ductility and formability.

2. State the main effects of Mn and Si in steel.

Answer guide: Mn increases strength and hardenability and combines with sulfur as MnS; Si is a deoxidizer and strengthens ferrite, and in higher amount can improve spring/electrical properties.

3. How do Mo and V improve alloy steels?

Answer guide: Mo improves hardenability, high-temperature strength and tempering resistance; V forms hard carbides/nitrides, refines grain size and increases wear resistance/strength.

4. Differentiate stainless steel and tool steel based on service requirement.

Answer guide: Stainless steel is selected mainly for corrosion/oxidation resistance; tool steel is selected for hardness, wear resistance and hot hardness in cutting/forming tools.

5. Why can HSLA steel reduce the mass of an automobile component?

Answer guide: Its higher yield strength allows a thinner section for the same load; it also retains useful toughness and weldability; therefore component mass can be reduced without using a low-density metal.

6. What is special about strengthening in maraging steel?

Answer guide: It has very low carbon and high Ni; martensite is relatively tough; aging forms fine intermetallic precipitates that produce very high strength.

7. Compare the graphite shape in grey and spheroidal graphite cast irons.

Answer guide: Grey iron has graphite flakes that create stress concentration; SG iron has graphite nodules/spheres that reduce stress concentration and improve strength and ductility.

8. Why is white cast iron more wear resistant but more brittle than grey cast iron?

Answer guide: Carbon is mainly present as hard cementite rather than graphite; cementite gives high hardness/wear resistance but low ductility and toughness.

9. Give one typical application each for brass, bronze and cupronickel with reason.

Answer guide: Brass: radiator/fittings - formable and corrosion resistant; bronze: bearing bush - wear resistant; cupronickel: condenser tube - seawater corrosion resistant.

10. Explain the purpose of quenching in Al-Cu precipitation hardening.

Answer guide: Rapid cooling suppresses equilibrium precipitation and retains Cu in a supersaturated solid solution; this prepares the alloy for controlled fine precipitation during aging.

11. Why are titanium alloys used in aerospace components?

Answer guide: They provide high strength-to-weight ratio, good corrosion resistance and useful elevated-temperature capability, although cost and machining are limitations.

12. Why are material standards important when purchasing engineering materials?

Answer guide: Standards provide common grade designation, composition/property limits, test methods and acceptance requirements, reducing ambiguity between suppliers and ensuring repeatability.

C. Twelve-Mark Questions with Answer Frameworks

1. Analyze the effects of Mn, Si, Cr, Mo, Ni, V, Ti and W on steels and relate them to engineering applications.

Suggested answer framework: Use a table: alloying element - principal effect - typical application. Include hardenability, toughness, corrosion resistance, carbide formation, grain refinement and hot hardness. Finish with 2-3 material-selection examples.

2. Compare stainless steels, tool steels, HSLA steels and maraging steels based on composition concept, important properties and applications.

Suggested answer framework: Begin with the service requirement of each class. Use a four-column comparison table covering strengthening/alloying concept, important properties and typical applications. Conclude with one component selection for each.

3. Compare grey, white, malleable and spheroidal graphite cast irons and select a suitable type for four different engineering components.

Suggested answer framework: Explain graphite/cementite morphology first. Then compare hardness, machinability, damping, ductility and strength. Use applications such as machine bed, crusher roll, pipe fitting and crankshaft.

4. Compare brass, bronze and cupronickel and explain how composition controls their applications.

Suggested answer framework: State the base compositions Cu-Zn, Cu-Sn and Cu-Ni. Compare corrosion resistance, wear resistance, formability and marine performance; then map each alloy to an application.

5. Explain precipitation strengthening of Al-Cu alloys with a clear process sequence and microstructural changes.

Suggested answer framework: Write flowchart: solution treatment -> quench -> aging. Explain supersaturated solid solution, fine coherent/semi-coherent precipitates, dislocation obstruction, peak aging and overaging.

6. Discuss the properties and applications of aluminium, magnesium and titanium alloys for lightweight engineering design.

Suggested answer framework: Compare density, strength-to-weight ratio, corrosion behavior, temperature capability, manufacturability and cost. Give automobile/aerospace examples and explain selection trade-offs.

7. Explain why Ni-based superalloys are used in gas-turbine hot sections and discuss the role of creep resistance.

Suggested answer framework: Describe high-temperature strength, oxidation/corrosion resistance and stable strengthening phases. Relate to turbine blades/discs and explain why ordinary steels lose strength at such temperatures.

8. Explain shape memory effect and superelasticity in shape memory alloys with suitable applications.

Suggested answer framework: Introduce reversible martensitic transformation. Explain deformed martensite -> heating -> austenite shape recovery and stress-induced martensite for superelasticity. Add actuator, coupling and biomedical examples.

Unit III Rapid Revision Map

Engineering need	First material to recall	Key reason
Corrosion-resistant tank	Stainless steel	Chromium passivation
Hot cutting edge	Tool steel	Hard carbides and hot hardness
Strong weldable auto structure	HSLA steel	High yield strength with useful weldability
Very-high-strength steel component	Maraging steel	Age-hardened intermetallic precipitates
Machine bed	Grey cast iron	Damping and machinability
Crusher roll	White cast iron	Cementite - very high wear resistance
Seawater condenser tube	Cupronickel	Marine corrosion/erosion resistance
Gas-turbine hot section	Ni-based superalloy	High-temperature strength and creep resistance
Thermal actuator	Shape memory alloy	Shape recovery on heating

UNIT IV - NON-METALLIC MATERIALS: QUESTION BANK

Coverage: commodity and engineering polymers; thermoplastics and thermosets; engineering ceramics; intermetallics; composites; matrix and reinforcement; applications; nanocomposites.

A. One-Mark / MCQ Practice

No.	Question	Ans.
1	A low-cost flexible packaging film is required. Which polymer is commonly suitable? (a) PE (b) PEEK (c) PAI (d) SIALON	a
2	A low-density food container with good chemical resistance is required. Which commodity polymer is suitable? (a) PP (b) PSZ (c) Al ₂ O ₃ (d) Bronze	a
3	A transparent but brittle disposable product is needed at low cost. Which polymer is a common choice? (a) PS (b) PEEK (c) PTFE (d) PAI	a
4	A transparent sheet with good weather resistance is needed, but extremely high impact resistance is not essential. Which polymer is appropriate? (a) PMMA (b) PTFE (c) PPS (d) Phenolic	a
5	A transparent impact-resistant machine guard is required. Which polymer is the best choice? (a) PC (b) PMMA (c) PS (d) UF	a
6	A light engineering gear needs good wear resistance and low friction. Which polymer is commonly used? (a) PA (Nylon) (b) PVC (c) PS (d) Urea formaldehyde	a
7	A component works continuously at high temperature and in aggressive chemicals. Which engineering polymer is suitable? (a) PEEK (b) PE (c) PS (d) PVC	a

No.	Question	Ans.
8	A seal requires extremely low friction and outstanding chemical inertness. Which polymer is preferred? (a) PTFE (b) PET (c) ABS (d) PP	a
9	Which thermoset is widely used for electrical switches, handles and heat-resistant molded parts? (a) Phenol formaldehyde (b) PE (c) PP (d) PET	a
10	Which ceramic is widely used as an electrical insulator and wear-resistant substrate? (a) Al ₂ O ₃ (b) PE (c) Brass (d) PVC	a
11	A high-speed rolling bearing needs low density, good strength and thermal shock resistance. Which ceramic is suitable? (a) Si ₃ N ₄ (b) Al ₂ O ₃ only (c) PS (d) PMMA	a
12	Which ceramic obtains improved toughness through stress-induced transformation? (a) PSZ (b) SiC (c) Al ₂ O ₃ (d) SIALON	a
13	In a fibre-reinforced composite, which constituent mainly provides stiffness and strength? (a) Reinforcement (b) Matrix (c) Pigment (d) Plasticizer	a
14	What is a key function of the matrix in a fibre composite? (a) Bind and protect reinforcement and transfer load (b) Always carry all the load (c) Melt reinforcement (d) Eliminate interfaces	a
15	A polymer containing nanoscale clay or graphene to improve barrier/stiffness is called a: (a) Nanocomposite (b) Intermetallic (c) Cast iron (d) Solid solution	a

Quick reason key:

1. a - Polyethylene is low cost, flexible, chemically resistant and widely used for films.

2. a - Polypropylene is lightweight, chemical resistant and widely used for containers.
3. a - Polystyrene is clear and low cost but relatively brittle.
4. a - PMMA has high optical clarity and good weather resistance.
5. a - Polycarbonate has very high impact resistance with good transparency.
6. a - Nylon combines strength, wear resistance and low friction.
7. a - PEEK has high temperature capability, strength and chemical resistance.
8. a - PTFE has very low coefficient of friction and excellent chemical resistance.
9. a - Phenolic resins are rigid, heat resistant and good electrical insulators.
10. a - Alumina is hard, chemically stable and electrically insulating.
11. a - Silicon nitride is used for high-speed ceramic bearing elements.
12. a - Partially stabilized zirconia uses transformation toughening.
13. a - Fibres/particles carry a major portion of load and provide stiffness/strength.
14. a - The matrix holds fibres, protects them and transfers load through the interface.
15. a - Nanocomposites contain at least one reinforcing feature in the nanoscale range.

B. Three-Mark Questions with Answer Guides

1. Differentiate commodity polymers and engineering polymers with two examples each.

Answer guide: Commodity polymers are low-cost, high-volume materials for general use (PE, PP, PS, PVC); engineering polymers offer higher strength/temperature/wear performance for functional parts (PC, PA, PEEK, PPS, PAI).

2. Compare PE and PP for common engineering products.

Answer guide: PE is generally more flexible and widely used in films/pipes; PP is stiffer and has better heat resistance, useful for containers, hinges and automotive trim; both have good chemical resistance and low density.

3. Why is PC preferred to PMMA for an impact-resistant transparent guard?

Answer guide: Both are transparent, but PC has much higher impact strength; PMMA has excellent clarity/weather resistance but is more brittle; therefore PC is safer for impact loading.

4. Why is nylon (PA) used for gears and bushes?

Answer guide: Nylon has good wear resistance, low friction, adequate strength and low mass; it also reduces noise, though moisture absorption must be considered.

5. State the important properties of PEEK and one application.

Answer guide: High continuous-use temperature, good mechanical strength, fatigue/chemical resistance and dimensional stability; used in aerospace, chemical, medical or high-performance bearing components.

6. Differentiate thermoplastics and thermosets using molecular structure and heating behavior.

Answer guide: Thermoplastics have mainly linear/branched chains and can soften repeatedly on heating; thermosets have a cross-linked network and do not remelt after curing.

7. Compare urea formaldehyde and phenol formaldehyde in simple terms.

Answer guide: Both are thermosets. UF is hard, rigid and good for electrical/adhesive uses but has lower heat/moisture resistance; phenolic has better heat resistance, dimensional stability and electrical insulation.

8. Why are Al₂O₃ and SiC used for wear/abrasive applications?

Answer guide: Both are very hard and chemically stable; alumina is also a good electrical insulator, while SiC has high thermal conductivity and excellent abrasion/high-temperature performance.

9. Why is Si₃N₄ preferred for some high-speed bearings?

Answer guide: It has low density, high hardness, good fracture resistance, low thermal expansion and good thermal-shock behavior, reducing centrifugal load and improving bearing performance.

10. Explain transformation toughening in PSZ.

Answer guide: A stressed crack tip triggers tetragonal-to-monoclinic transformation; the associated local volume expansion creates compressive stress around the crack and resists crack growth.

11. State the functions of matrix and reinforcement in a composite.

Answer guide: Matrix binds/protects reinforcement and transfers load; reinforcement provides most of the stiffness/strength and controls direction-dependent mechanical performance.

12. Why can nanocomposites show large property improvements with small reinforcement content?

Answer guide: Nanofillers have very high surface area and strong interfacial interaction; well-dispersed particles can restrict polymer-chain motion, improve load transfer and create a tortuous path for gas/moisture.

C. Twelve-Mark Questions with Answer Frameworks

1. Classify polymers into commodity, engineering, thermoplastic and thermoset categories and explain their engineering significance with examples.

Suggested answer framework: Start with classification. Use a table listing examples from the syllabus, key properties and applications. Clearly distinguish reheatable thermoplastics from cross-linked thermosets.

2. Compare the properties and applications of PE, PP, PS, PVC, PMMA and PET.

Suggested answer framework: Use a six-row table covering transparency/flexibility, chemical resistance, strength, temperature behavior and common products. Finish with material-selection examples.

3. Compare PC, PA, ABS, PAI, PPO, PPS, PEEK and PTFE as engineering polymers.

Suggested answer framework: Organize by key advantage: impact, wear, toughness, high temperature, dimensional stability, chemical resistance and low friction. Give one engineering application for each.

4. Explain the properties and applications of Al₂O₃, SiC, Si₃N₄, PSZ and SIALON.

Suggested answer framework: Use a comparison table. Include hardness, density, toughness, thermal shock, insulation/conductivity and high-temperature behavior. Map to insulators, abrasives, bearings, biomedical/tough components and cutting tools.

5. Explain intermetallic compounds and discuss why they are attractive but difficult for high-temperature structural applications.

Suggested answer framework: Define ordered compounds such as TiAl/NiAl. Mention low density or high-temperature strength/oxidation resistance; also discuss room-temperature brittleness and manufacturing difficulty.

6. Explain the constituents of a composite, load transfer and factors controlling composite properties.

Suggested answer framework: Define matrix, reinforcement and interface. Explain fibre load carrying and matrix shear transfer. Discuss fibre type, volume fraction, orientation, aspect ratio and interfacial bonding.

7. Discuss major applications of composites in aerospace, automobile, wind energy and sports engineering.

Suggested answer framework: For each sector, name specific components and link benefits such as low weight, corrosion resistance, high specific stiffness and fatigue performance to the application.

8. Explain nanocomposites, their strengthening concept, advantages, limitations and engineering applications.

Suggested answer framework: Define nanoscale reinforcement. Discuss high surface area, interfacial effects, load transfer and barrier path. Include improved stiffness/wear/barrier properties plus agglomeration, processing and cost limitations.

AT-2 MODEL QUESTIONS AND ANSWERS - STUDENT COPY

Note: The following questions match the AT-2 question paper. The answers are written as model answers.

Equivalent technically correct wording is acceptable. Students should learn the concept and presentation rather than memorizing sentences.

PART A - 10 x 1 = 10 Marks

Q.No	Question	Correct answer	Why?
1	A stainless-steel vessel must resist rusting during service. Which alloying element is most directly responsible for this corrosion resistance?	(b) Cr	Chromium forms a thin, stable passive oxide film on stainless steel and greatly improves corrosion resistance.
2	A cutting tool must retain high hardness even when the cutting edge becomes hot. Which steel class is the most suitable?	(b) Tool steel	Tool steels are designed for high hardness, wear resistance and, in many grades, hot hardness during cutting.
3	A seawater condenser tube requires good resistance to marine corrosion. Select the most suitable copper alloy.	(c) Cupronickel	Cu-Ni alloys have excellent resistance to seawater corrosion and erosion, making them suitable for condenser and marine tubes.
4	An Al-Cu alloy has been solution treated and quenched. Which next step develops fine precipitates and increases its strength?	(b) Aging	Aging allows fine precipitates to form from the supersaturated solid solution. These precipitates hinder dislocation motion and increase strength.
5	A deformed component returns close to its original shape when heated. Which material is best suited for this behaviour?	(b) Shape memory alloy	Shape memory alloys recover their original shape on heating because of a reversible martensitic transformation.
6	A transparent machine guard must withstand accidental impact. Which polymer is the better choice?	(c) PC	Polycarbonate combines transparency with very high impact resistance.
7	A seal requires very low friction and high chemical resistance. Which polymer is most suitable?	(b) PTFE	PTFE has an extremely low coefficient of friction and excellent chemical inertness.
8	A high-speed bearing needs a ceramic with low density, good strength and thermal-shock resistance. Select the suitable material.	(a) Si₃N₄	Silicon nitride is low in density, hard, strong and resistant to thermal shock; it is widely used in advanced bearing applications.
9	In a fibre-reinforced composite, which part mainly carries the applied load?	(b) Reinforcement	The reinforcement, especially fibres, supplies most of the stiffness and strength and carries a major part of the load.
10	A polymer is modified by adding nanosized particles to improve stiffness and barrier properties. The resulting material is called:	(b) Nanocomposite	A nanocomposite contains nanoscale reinforcement dispersed in a matrix to improve selected properties.

PART B - 10 x 3 = 30 Marks

How to write a 3-mark answer
Start by giving the material/process selection directly.
Write 2-3 technical reasons in separate points.
Avoid one long paragraph; short point-wise answers are easier to evaluate.

11. A gear steel needs good hardenability and toughness. Analyze how chromium and nickel help to meet these requirements.

Model answer:

- Chromium increases hardenability, so a deeper section can transform to a hard microstructure during heat treatment.
- Chromium also forms hard carbides and improves wear resistance.
- Nickel improves toughness and ductility and helps the steel resist shock loading. Therefore Cr-Ni steel gives a useful combination of strength/hardness and toughness.

12. A crusher roll requires high wear resistance. Select a suitable type of cast iron and justify the selection using its microstructure and property.

Model answer:

- Suitable material: White cast iron.
- In white cast iron, much of the carbon is present as hard iron carbide (cementite) rather than free graphite.
- The cementite-rich structure gives very high hardness and abrasion resistance, which is suitable for crusher rolls. Its brittleness is the main limitation.

13. A heat-exchanger tube will operate in seawater. Select brass, bronze or cupronickel and justify your choice using two relevant properties.

Model answer:

- Suitable material: Cupronickel (Cu-Ni alloy).
- It has very good resistance to seawater corrosion and erosion by flowing water.
- It also has adequate strength and good resistance to marine service conditions, so it is widely used for condenser/heat-exchanger tubes.

14. An Al-Cu alloy is solution treated and quenched but is still relatively soft. Explain why an aging treatment is required to increase its strength.

Model answer:

- After quenching, Cu is trapped in a supersaturated aluminium solid solution.
- During aging, very fine Cu-rich precipitates form throughout the matrix.
- These fine precipitates obstruct dislocation movement. More stress is then required for plastic deformation, so hardness and strength increase.

15. Two suppliers offer the same alloy under different trade names. Explain why a recognized material standard should be checked before accepting the material. Mention any two items that a standard may specify.

Model answer:

- A recognized standard gives a common technical definition of a material grade and avoids confusion caused by trade names.
- It helps the buyer verify that material from different suppliers meets the same required quality.
- A standard may specify chemical composition, mechanical-property limits, test methods, heat-treatment condition, dimensions/tolerances or grade designation. Any two relevant items can be written.

16. A transparent safety cover must resist impact. Compare PC and PMMA for this requirement and select the more suitable polymer.

Model answer:

- PMMA has excellent transparency and weather resistance but is comparatively brittle.
- PC is also transparent and has much higher impact strength and toughness.
- Therefore polycarbonate (PC) is the better choice for an impact-resistant safety cover.

17. A plastic part must be reshaped by heating, while an electrical switch body must retain its shape when heated. Relate these requirements to thermoplastics and thermosets.

Model answer:

- The reshaped part should be a thermoplastic because thermoplastics soften on heating and can be remoulded.
- The electrical switch body should be a thermoset because the cross-linked structure does not melt again after curing.
- Thus thermoplastics are suitable for repeated shaping, while thermosets are suitable where heat resistance and dimensional stability are needed.

18. A high-speed bearing operates under repeated loading and temperature changes. Select Si₃N₄ or Al₂O₃ and justify the choice using suitable properties.

Model answer:

- Preferred material: Si₃N₄.
- Si₃N₄ has lower density, good fracture resistance and good thermal-shock resistance compared with alumina.
- Its low density reduces centrifugal loading at high speed, while its hardness and fatigue capability are useful for bearing elements.

19. In a fibre-reinforced composite, explain the functions of the matrix and reinforcement using one engineering example.

Model answer:

- Matrix: binds the fibres, gives shape, protects them from damage/environment and transfers load to the fibres through the interface.
- Reinforcement: carries a major portion of the load and provides high strength and stiffness.
- Example: In glass-fibre reinforced polymer (GFRP), epoxy/polymer is the matrix and glass fibres are the reinforcement; it is used in automobile panels, leaf springs or wind-blade structures.

20. Nanoparticles are added in small quantity to a polymer matrix. Analyze why they can improve the material properties and state one engineering application.

Model answer:

- Nanoparticles have very high surface area, so a large interface is created even at a small amount of reinforcement.
- If they are well dispersed, they restrict polymer-chain movement, improve load transfer and can increase stiffness, strength, wear or barrier performance.
- Example: polymer-clay nanocomposite for packaging/automotive panels, or nano-silica/graphene modified polymer for wear-resistant/lightweight components.

PART C - 5 x 12 = 60 Marks

How to write a 12-mark answer
Use a short introduction and then organize the main answer under clear subheadings.
For material-selection questions, use a comparison/selection table. For treatment questions, start with a process flowchart.
Connect every selected material to a required property; do not only list applications.
A neat table plus short explanations is better than one long paragraph.

21(a). An engineering company needs steels for: (i) a corrosion-resistant chemical tank, (ii) a cutting tool, (iii) a lightweight automobile structural member, and (iv) a very-high-strength aerospace component. Select a suitable material from stainless steel, tool steel, HSLA steel and maraging steel for each application. Justify each selection using the required properties and the role of important alloying elements.

Model answer

Different classes of steels are designed by controlling alloy content and strengthening mechanism. The correct steel is selected by matching its main property to the service requirement.

Component / requirement	Selected steel	Important properties / alloying reason
Chemical tank - corrosion resistance	Stainless steel	Cr (normally at least about 10.5%) forms a passive oxide film; Ni in many grades improves toughness/formability and stabilizes austenite.
Cutting tool - high hardness and wear resistance	Tool steel	C with Cr/W/Mo/V forms hard carbides; W/Mo improve hot hardness and temper resistance; V improves wear resistance and grain refinement.
Automobile structural member - high strength with reduced section	HSLA steel	Small additions such as Mn, V, Nb/Ti increase yield strength through grain refinement/precipitation while maintaining useful weldability. Higher strength allows thinner sections and lower component mass.
Aerospace/high-strength component	Maraging steel	Very low C, high Ni steel. Soft/tough martensitic matrix is aged to form fine intermetallic precipitates, giving extremely high strength with good toughness.

- Conclusion: stainless steel is selected for corrosion resistance, tool steel for cutting performance, HSLA for strong weldable lightweight structural design, and maraging steel for exceptional strength and toughness.

21(b). A steel designer can add Mn, Si, Cr, Mo, Ni, V, Ti and W. Analyze the main effect of each alloying element on steel. Then recommend suitable alloying additions for: (i) a corrosion-resistant component, (ii) a tough shaft, and (iii) a high-temperature cutting tool, with reasons.

Model answer

Element	Main effect in steel
Mn	Increases strength and hardenability; deoxidizer; combines with sulfur as MnS and reduces hot-shortness.
Si	Deoxidizer; strengthens ferrite; useful in spring/electrical steels at higher content.
Cr	Improves hardenability, wear and oxidation/corrosion resistance; forms hard carbides.
Mo	Improves hardenability, high-temperature strength and tempering resistance; helps resist temper brittleness.
Ni	Improves toughness, ductility and corrosion resistance; useful for low-temperature toughness.
V	Forms hard carbides/nitrides, refines grains, increases strength and wear resistance.
Ti	Strong carbide/nitride former; grain refinement; stabilizes C/N and is used to control intergranular corrosion in stabilized stainless grades.
W	Forms stable hard carbides and gives hot/red hardness in tool and high-speed steels.

Recommended additions

- Corrosion-resistant component: Cr is essential; Ni may be added to improve toughness and stabilize austenite. This is the basis of many stainless steels.
- Tough shaft: Ni for toughness together with Cr/Mo for hardenability and strength is a common combination.
- High-temperature cutting tool: W or Mo with V/Cr because stable carbides maintain hardness and wear resistance at elevated cutting temperatures.

22(a). A design team must select materials for: (i) an electrical connector, (ii) a marine condenser tube, (iii) a lightweight aircraft structural part, (iv) a high-temperature turbine blade, and (v) a compact thermal

actuator. Select suitable materials from copper/copper alloys, aluminium alloys, titanium alloys, nickel-based superalloys and shape memory alloys. Justify the selection for each application using relevant properties.

Model answer

Application	Suitable material	Reason for selection
Electrical connector	Copper / suitable Cu alloy	Very high electrical conductivity, good ductility and adequate corrosion resistance.
Marine condenser tube	Cupronickel	Excellent seawater corrosion and erosion resistance with adequate strength.
Lightweight aircraft structural part	Aluminium alloy	Low density, good strength-to-weight ratio, formability and corrosion resistance.
High-temperature turbine blade	Ni-based superalloy	Retains strength at high temperature, excellent creep resistance and oxidation/hot-corrosion resistance.
Compact thermal actuator	Shape memory alloy (e.g., NiTi concept)	Produces controlled shape recovery on heating and can generate useful actuation force in a compact size.

- Titanium alloys are also important in aerospace where higher specific strength, corrosion resistance and temperature capability are needed, but they are more expensive than aluminium alloys.

22(b). A manufacturer is choosing copper alloys for three components: (i) a radiator tube, (ii) a bearing bush, and (iii) a marine condenser tube. Compare brass, bronze and cupronickel in terms of composition, important properties and applications, and select a suitable alloy for each component with justification.

Model answer

Alloy	Basic composition	Important properties	Typical applications
Brass	Cu-Zn	Good formability, corrosion resistance, machinability and useful thermal conductivity	Radiator tubes, fittings, screws, decorative parts
Bronze	Commonly Cu-Sn (many bronze families exist)	Good wear resistance, strength, corrosion resistance and anti-friction behavior	Bearing bushes, gears, marine fittings
Cupronickel	Cu-Ni	Excellent seawater corrosion/erosion resistance and good strength	Marine condenser and heat-exchanger tubes, seawater piping

Selection

- Radiator tube: Brass - easy to form into tubes and has useful corrosion resistance and thermal performance.
- Bearing bush: Bronze - good wear resistance and anti-friction/load-bearing behavior.
- Marine condenser tube: Cupronickel - superior resistance to seawater corrosion and erosion.

23(a). An Al-Cu alloy must be strengthened by precipitation hardening. Develop the complete treatment sequence from solution treatment to aging. Explain the microstructural changes during each stage and analyze the effects of under-aging, peak-aging and over-aging on strength.

Model answer

Process sequence: Solution treatment -> Rapid quenching -> Aging (natural or artificial)

1. Solution treatment

- Heat the Al-Cu alloy into the single-phase alpha region above the solvus temperature but below the solidus.
- Cu-rich second-phase particles dissolve and Cu atoms enter the aluminium solid solution.

2. Quenching

- Cool rapidly to room temperature.
- Diffusion is suppressed, so Cu atoms are retained in a supersaturated solid solution. The alloy is not yet at maximum strength.

3. Aging

- Hold at room temperature (natural aging) or at an elevated temperature (artificial aging).
- Very fine Cu-rich zones/precipitates form. They create strain fields and obstacles that hinder dislocation movement, so strength and hardness increase.

Condition	Microstructural condition	Effect on strength
Under-aged	Precipitates are still developing; spacing/size not yet optimum	Strength increasing but below maximum
Peak-aged	Fine, closely spaced strengthening precipitates give maximum resistance to dislocation motion	Maximum or near-maximum strength/hardness
Over-aged	Precipitates coarsen and become more widely spaced / less effective barriers	Strength and hardness decrease; ductility may increase

- Exam diagram suggestion: Draw a simple strength/hardness versus aging time curve showing under-aged, peak-aged and over-aged regions.

23(b). A foundry must choose cast irons for: (i) a machine bed, (ii) a crushing roll, (iii) a pipe fitting, and (iv) an automobile crankshaft. Select from grey, white, malleable and spheroidal graphite cast irons. Explain the important microstructural feature and justify the choice for each application.

Model answer

Component	Cast iron selected	Microstructural feature	Why suitable
Machine bed	Grey cast iron	Graphite flakes in ferritic/pearlitic matrix	Excellent vibration damping, machinability and good compressive strength.
Crushing roll	White cast iron	Carbon mainly as hard cementite/carbides; little or no free graphite	Very high hardness and abrasion resistance.
Pipe fitting	Malleable cast iron	Temper-carbon nodules/clusters formed by annealing white iron	Better ductility and toughness than white/grey iron; suitable for fittings and shock loading.
Automobile crankshaft	Spheroidal graphite (ductile) iron	Graphite present as nearly spherical nodules	High strength, fatigue resistance and ductility because nodules cause less stress concentration than flakes.

- Key concept: graphite shape strongly affects mechanical behavior. Flakes concentrate stress, while spheroidal nodules greatly reduce stress concentration.

24(a). A product-design team needs polymers for: (i) a soft packaging film, (ii) a transparent beverage bottle, (iii) a transparent impact-resistant machine guard, (iv) a lightweight gear, and (v) a high-temperature chemical component. Select suitable materials from PE, PET, PC, PA (Nylon), PEEK and related polymers in the syllabus. Justify each choice using the required properties.

Model answer

Requirement	Selected polymer	Reason
Soft packaging film	PE	Flexible, low cost, lightweight, moisture/chemical resistant and easily processed into film.
Transparent beverage bottle	PET	Transparent, good strength and toughness, good gas/barrier performance and widely blow moulded into bottles.
Impact-resistant transparent machine guard	PC	Very high impact strength combined with transparency and dimensional stability.
Lightweight gear	PA (Nylon)	Good wear resistance, low friction, adequate strength, low mass and quiet operation.
High-temperature chemical component	PEEK	High service temperature, strong chemical resistance, good mechanical and fatigue

Requirement	Selected polymer	Reason
		properties.

- Other syllabus polymers are selected when their special property is needed: PTFE for very low friction/chemical inertness, PPS for chemical and heat resistance, PPO for dimensional stability/electrical uses, ABS for tough housings.

24(b). A manufacturer needs materials for: (i) a low-cost food container, (ii) an electrical switch body exposed to heat, (iii) a wear-resistant small gear, and (iv) a low-friction chemical seal. Select suitable materials from PP, phenol formaldehyde, nylon (PA), PEEK and PTFE. Classify each selected polymer as thermoplastic or thermoset and justify the selection based on properties.

Model answer

Application	Material	Type	Reason
Low-cost food container	PP	Thermoplastic	Low density, good chemical resistance, adequate stiffness, low cost and easy moulding.
Electrical switch body exposed to heat	Phenol formaldehyde (phenolic)	Thermoset	Good electrical insulation, heat resistance, rigidity and dimensional stability after curing.
Wear-resistant small gear	Nylon (PA)	Thermoplastic	Good wear resistance, low friction, sufficient strength and low noise/weight.
Low-friction chemical seal	PTFE	Thermoplastic	Extremely low friction and outstanding chemical inertness.

- PEEK is an engineering thermoplastic and can be selected when the component requires a combination of high temperature, high strength and chemical resistance, but it is much more expensive than common polymers.

25(a). An engineering company needs ceramics for: (i) an electrical insulator, (ii) a wear-resistant abrasive component, (iii) a high-speed bearing, (iv) a tough biomedical/impact-resistant component, and (v) a high-temperature cutting component. Select suitable materials from Al₂O₃, SiC, Si₃N₄, PSZ and SiALON. Justify each selection using the relevant properties.

Model answer

Application	Ceramic selected	Key properties / reason
Electrical insulator	Al ₂ O ₃ (alumina)	Excellent electrical insulation, hardness, wear resistance and chemical stability.
Wear-resistant abrasive component	SiC	Very high hardness, excellent abrasion resistance, high thermal conductivity and high-temperature stability.
High-speed bearing	Si ₃ N ₄	Low density, high hardness, good fracture resistance, thermal-shock resistance and lower centrifugal load at speed.
Tough biomedical/impact-resistant component	PSZ	Transformation toughening gives unusually high fracture toughness for a ceramic; zirconia-based ceramics are also used in biomedical components.
High-temperature cutting component	SiALON	Good hot hardness, wear resistance, thermal-shock resistance and chemical stability at elevated temperature.

- Key comparison: ceramics are generally very hard and heat resistant, but their brittleness varies. Si₃N₄ and PSZ are selected where improved fracture/thermal-shock resistance is important.

25(b). An automobile company wants to replace a steel leaf spring with a lighter composite. Propose a suitable matrix and reinforcement, explain the function of each and how load is transferred through the

composite. Discuss the expected advantages and limitations. Finally, explain how adding nanosized reinforcement would convert it into a nanocomposite and what additional improvement may be expected.

Model answer

A suitable choice is a glass-fibre reinforced polymer (GFRP) leaf spring, for example an epoxy polymer matrix reinforced with continuous E-glass fibres aligned mainly along the spring length.

Constituents and functions

Constituent	Function
Polymer/epoxy matrix	Binds fibres, gives shape, protects them from moisture/abrasion and transfers load to the fibres through interfacial shear.
Glass fibres	Carry most of the tensile load and provide high specific strength and stiffness.
Interface	Provides bonding so stress can be transferred efficiently from matrix to fibres.

Load transfer

- When the leaf spring bends, the matrix first receives the applied load and transfers it by shear at the interface to the aligned fibres.
- The fibres carry the major tensile/compressive stresses because they are much stiffer and stronger than the polymer matrix.

Expected advantages over steel

- Large reduction in component mass because the composite has low density and high specific strength.
- Good corrosion resistance and useful fatigue performance.
- Possibility of integrating several steel leaves into a simpler composite component and reducing noise/vibration.

Limitations

- Higher material/manufacturing cost than ordinary steel in many cases.
- Anisotropic behavior: properties depend strongly on fibre orientation.
- Damage such as delamination can be difficult to detect and repair; joining and recycling are also more challenging.

Conversion to a nanocomposite

- Add a small amount of nanosized reinforcement such as nano-silica, nanoclay, graphene or carbon nanotubes to the polymer matrix, with good dispersion.
- The resulting multiscale material becomes a nanocomposite because one reinforcement scale is in the nanometre range.
- Possible additional benefits include improved matrix stiffness, wear resistance, crack resistance, barrier properties and interfacial performance. Poor dispersion/agglomeration can reduce these benefits.

Final revision reminder: For AT-2, focus on material selection based on required properties. Do not memorize only the material name; always learn the engineering reason for the selection.